AP CALCULUS AB

Unit 1 • Worksheet 1-9A

Worksheet 1-9A • IVT, EVT, and MVT

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

"Was there a time when the water level was exactly 3 m?" Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) none

Question 2

NO CALCULATOR

"Was the water level ever rising at exactly 0.5 m per hour?" Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) none

Question 3

NO CALCULATOR

The graph of f on $[0,4]$ has a corner at $x = 2$ and no breaks. Which theorems are guaranteed to apply on $[0,4]$?

(A) IVT only
(B) IVT and EVT
(C) EVT and MVT
(D) IVT, EVT and MVT

Question 4 (Free response • AP style)

NO CALCULATOR

A differentiable function f has the values below.

x	0	2	5	7	10
$f(x)$	4	-2	3	9	5

(a) Must $f(c) = 0$ for some c in $(0,2)$? Justify.
(b) Must $f'(c) = 3$ for some c in $(5,7)$? Justify.
(c) Must f attain an absolute minimum on $[0,10]$? Justify.
(d) Must $f'(c) = 0$ for some c in $(0,10)$? Justify.

Question 5 (Free response) NO CALCULATOR

Let $h(x) = x + 1$ for $0 \le x \le 2$ and $h(x) = 3 - \frac{1}{2}(x-2)^2$ for $2 < x \le 4$. Near $x = 2$, the left piece has slope 1 and the right piece has slope 0.

(a) Is h continuous at $x = 2$? Is it differentiable at $x = 2$? Justify.
(b) Which of the IVT, EVT and MVT apply to h on $[0, 4]$? Explain.
(c) Does the MVT apply on $[2, 4]$? If so, what is the guaranteed value of $h'(c)$?